

Material Notes: Solid Rigid Polyurethane Foam

Primary Use

Solid rigid polyurethane foam is primarily used as an alternative test medium for human cancellous bone. These products are not intended to replicate the mechanical properties of human bone, however, it does provide consistent and uniform material with properties in the range of human cancellous bone. Relevant mechanical properties for comparison to human cancellous bone may depend on the particular test method that is being developed.

The ASTM F-1839-08 "Standard Specification for Rigid Polyurethane Foam for Use as a Standard Material for Testing Orthopaedic Devices and Instruments" states that, "The uniformity and consistent properties of rigid polyurethane foam make it an ideal material for comparative testing of bones screws and other medical devices and instruments."

Description

Solid rigid polyurethane foam has a closed cell content ranging from 96.0 to 99.9%. Foam is available in a range of sizes and densities, from 0.08 to 0.80 grams per cubic centimeter (5 to 50 pounds per cubic foot).

Standard block size: 13cm X 18cm X 4cm
Standard sheet size: 13cm X 18cm X 3mm

Other Notes

Water absorption ranges from 0.301 to 0.0 kg/m²
Foam meets ASTM F-1839-08. Block dimensions are within ±2mm and sheet thickness is within ±0.3mm. Foam density may vary ±10%.
Properties parallel to direction of the rise of foam.

Information provided by Sawbones.

Vendors: No vendors are listed for this material. Please [click here](#) if you are a supplier and would like information on how to add your listing to this material.

Physical Properties	Metric	English	Comments
Density	0.160 g/cc	0.00578 lb/in ³	
Mechanical Properties	Metric	English	Comments
Tensile Strength	2.10 MPa	305 psi	
Tensile Modulus	0.0860 GPa	12.5 ksi	
Compressive Yield Strength	2.20 MPa	319 psi	
Compressive Modulus	0.0580 GPa	8.41 ksi	
Poissons Ratio	0.30	0.30	Calculated
Shear Modulus	0.0190 GPa	2.76 ksi	
Shear Strength	1.60 MPa	232 psi	
Thermal Properties	Metric	English	Comments
CTE, linear	63.0 µm/m-°C	35.0 µin/in-°F	
@Temperature -46.0 - 93.0 °C @Temperature -50.8 - 199 °F			

Some of the values displayed above may have been converted from their original units and/or rounded in order to display the information in a consistent format. Users requiring more precise data for scientific or engineering calculations can click on the property value to see the original value as well as raw conversions to equivalent units. We advise that you only use the original value or one of its raw conversions in your calculations to minimize rounding error. We also ask that you refer to MatWeb's [terms of use](#) regarding this information. [Click here](#) to view all the property values for this datasheet as they were originally entered into MatWeb.